

Homework Set 6. Direct Synthesis of Feedback and Feedforward Loops

(Due on Tuesday, April 8, 2025)

1. (15 pt) One-parameter filter-based controller synthesis. [Seborg 12.2]

Consider the following transfer functions:

$$G_p = \frac{2(1-s)}{s}, \quad G_v = 2, \quad G_m = 1.$$

Design an IMC controller. Derive an expression for $G_c(s)$ and report it in a standard form (e.g., P, PI, or PID, or other common types of transfer functions).**2. (20 pt) Controller synthesis for non-minimum phase and time-delay systems.** [Kravaris 19.1]

Apply the one-parameter direct controller synthesis to the following two process transfer functions.

$$G_{p1}(s) = \frac{K(1 - \tau_0 s)}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)}$$

where $\tau_0, \tau_1, \tau_2, \tau_3 > 0$.

$$G_{p2}(s) = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)} e^{-\theta s}$$

where $\tau_1, \tau_2, \theta > 0$. Assume $G_v = G_m = 1$.**3. (15 pt) Controller synthesis under uncertain (imprecise) models.** [Kravaris 19.6, simplified]

A process has transfer function of the form:

$$G_p(s) = \frac{K}{(\tau s + 1)(\varepsilon s + 1)}$$

where $\tau > 0$, $\varepsilon > 0$ and ε is a constant smaller than τ . The engineer ignored the fast dynamics and modeled the system as $\tilde{G}_p(s) = K/(\tau s + 1)$ instead. The engineer then followed the direct synthesis approach to build a PI controller

$$G_c(s) = \frac{\tau}{K\lambda} \left(1 + \frac{1}{\tau s} \right)$$

to control it, what will be the range of values of the filter time constant λ that guarantees stability?Assume $G_v = G_m = 1$.**4. (25 pt) Controller synthesis versus empirical tuning rules.** [Seborg 12.12]Suppose that a process can be adequately modeled by an FOPTD model: $G_p(s) = Ke^{-\theta s}/(\tau s + 1)$, and assume $G_v = G_m = 1$.(a) Directly synthesize a controller with a closed-loop time constant $\tau_c = \tau$.

(b) Cohen and Coon (1953) reported the following tuning relations to PI controllers:

$$K_c = \frac{\tau}{K\theta} \left(0.9 + \frac{\theta}{12\tau} \right), \quad \tau_I = \frac{30 + 3(\theta/\tau)}{9 + 20(\theta/\tau)} \theta.$$

These tuning relations aimed to provide closed-loop responses with a 1/4 decay ratio. Compare the PI settings calculated from these equations to the controller settings of part (a). Which one is more *conservative*? You may do a simulation to check.

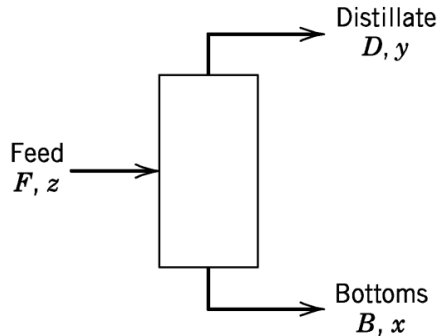
- (c) Let the model parameters be $K = 2$, $\tau = 3$, and $\theta = 1$, and simulate the controllers of parts (a) and (b) for a unit step change in set point. Which controller provides *better* control?

5. (25pt) Design of feedforward–feedback loops. [Seborg 15.10]

The distillation column in the figure below has the following transfer function model:

$$Y'(s) = \frac{2e^{-20s}}{95s + 1}D'(s) + \frac{0.5e^{-20s}}{45s + 1}F'(s)$$

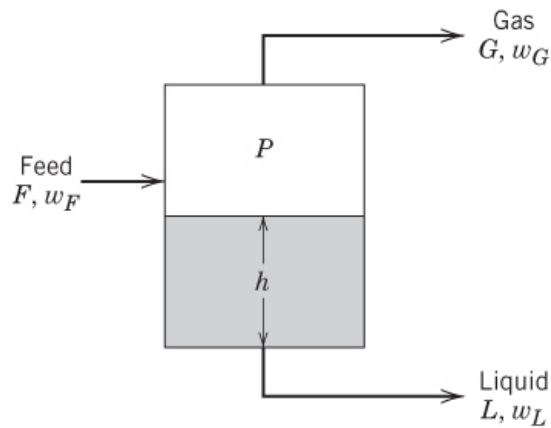
with $G_v = G_m = 1$. The distillate composition y is the controlled variable, the distillate flow rate D is the manipulated variable (the “ u ” in control theory), and the feed flow rate F is a disturbance (the “ d ” in control theory).



- (a) What is the feedforward controller that the engineer obtains?
[Hint: Since $G_p(s)$ and the model of $G_d(s)$ have the same delay, the delay factors should be exactly cancelled when designing $G_f(s)$. Such a cancellation would still make $G_f(s)$ proper.]
- (b) Synthesize a feedback controller with $\tau_c = 30$.
- (c) Simulate the closed-loop response of $y(t)$ to a unit step change in the disturbance using *feedback* control only.
- (d) Repeat part (c) for the *feedforward–feedback* control scheme.
- (e) However, the disturbance model $\frac{0.5e^{-20s}}{45s + 1}$ used for feedforward controller synthesis was erroneous. The actual disturbance effect (which is a hidden fact that the engineer could not know) should be $\frac{0.5e^{-30s}}{60s + 1}$. Repeat part (d) and describe what you observe.
[Caution: The actual $G_d(s)$ is different from the model that you previously used for feedforward controller synthesis. Hence, you should simulate with
- $$\frac{Y}{D} = \frac{G_d + G_f G_p}{1 + G_{OL}}$$
- where $G_d(s)$ is the actual one (instead of the model), and $G_f(s)$ is the one obtained in (a) based on the erroneous model.]

6. (Bonus: 10 pt) Process control in practice: Variable selection. [Seborg 13.10]

Whereas we often assume that the MVs and CVs are already defined before doing control problems, in fact, when engineers design each unit in a chemical plant, they need to determine which variables are MVs and which are CVs. This procedure is called *variable selection*. Please read Chapter 13 of Seborg and then answer the following question.



A two-phase feed to the gas-liquid separator (or flash drum), shown in the figure above, consists of a mixture of two hydrocarbons. Because the vessel pressure P is lower than the feed pressure, the feed flashes as it enters the separator. Using the following information and a degrees of freedom analysis, do the following:

- (a) Determine the model degrees of freedom, N_F .
- (b) Determine the control degrees of freedom, N_{FC} .
- (c) Specify reasonable CVs and MVs. Justify your answers.

Available Information:

1. The flash drum operates isothermally with the two phases in equilibrium.
2. Each phase is perfectly mixed.
3. The mass flow rates are in units of kg/min and the compositions (e.g., w_F) are expressed as mass fractions.
4. Each flow rate can be adjusted by a control valve (not shown).
5. For the uncontrolled process, the exit flow rates are related to vessel conditions by empirical equations that have the following forms: $G = f_1(P)$ and $L = f_2(h)$.

7. (Bonus: 10 pt) Process control in depth: Controller design for unstable plants. *You may or may not need to use Matlab. [Dorf Design Problem 9.10]*

Reactors often have unstable transfer functions (i.e., with right-half-plane poles). You may know about Chernobyl in 1986. Indeed, unstable plants are much harder to deal with the stable plants. The direct synthesis approach taught in this course (Lecture 16 and 17) assumes stable plants – which makes it not applicable to unstable processes. Now we have a reactor with an unstable model:

$$G_p(s) = \frac{4}{(s - 4)(s + 2)}, \quad G_v = G_m = 1.$$

It has an unstable pole (4) and a stable pole (−2). You are asked to design a controller $G_c(s)$ that

- (a) achieves closed-loop stability,
- (b) achieves zero offset (*Hint:* for this, $G_p(s)G_c(s)$ should contain a pole at $s = 0$), and also,
- (c) is not based on an exact pole-zero cancellation, i.e., $G_c(s)$ can not be designed to have a factor of exactly $(s - 4)$ in its numerator to “cancel” the pole at +4.

Under these constraints, try to minimize ISE + ISC under a step change in the setpoint.

Remark: An illustration of temperature control of nuclear reactors is shown in the following figure.

